

EXERCISE 4 JOINS, UNION, FILTERING and AGGREGATES

SQL JOINS

1. INNER JOIN

```
SELECT e.EmployeeID, e.FirstName, e.LastName, e.Department,
       e.Salary, p.ProjectID, p.ProjectName, p.Budget, p.Status
FROM Employees AS e
INNER JOIN Projects AS p
ON e.EmployeeID = p.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	Project Name	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk management	80000	Pending

2. LEFT JOIN

```
SELECT e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary,
       p.ProjectID, p.ProjectName, p.Budget, p.Status
FROM Employees AS e
LEFT JOIN Projects AS p
ON e.EmployeeID = p.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	Project Name	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk management	80000	Pending

3. RIGHT JOIN

```
SELECT p.ProjectID, p.ProjectName, p.Budget, p.Status, e.EmployeeID,
       e.FirstName, e.LastName, e.Department, e.Salary
FROM Employees AS e
RIGHT JOIN Projects AS p
ON e.EmployeeID = p.EmployeeID;
```

ProjectID	ProjectName	Budget	Status	EmpID	FirstName	LastName	Depart	Salary
101	AI development	100000	Completed	1	John	Doe	IT	70000
102	Employee Training	50000	Ongoing	2	Alice	Smith	HR	60000
103	Cybersecurity Audit	75000	Pending	1	John	Doe	IT	70000
104	Financial Analysis	90000	Ongoing	3	Bob	Johnson	Finance	75000
105	Market Expansion	65000	Completed	5	Emma	Wilson	Sales	65000
106	Risk management	80000	Pending	6	Michael	Clark	Finance	80000

4. FULL OUTER JOIN

```
SELECT e.EmployeeID, e.FirstName, e.LastName, e.Department, e.Salary,
       p.ProjectID, p.ProjectName, p.Budget, p.Status
FROM Employees AS e
FULL OUTER JOIN Projects AS p
ON e.EmployeeID = p.EmployeeID;
```

EmpID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk management	80000	Pending
1	John	Doe	IT	70000	101	AI Development	100000	Completed

5. UNION

SELECT City AS Location
FROM Employees

UNION

SELECT status AS Location
FROM Projects;

Location
New York
Los Angeles
Toronto
London
Sydney
Completed
Ongoing
Pending

6. SELECT City AS Location FROM Employees
UNION ALL

SELECT status AS Location FROM Projects;

Location
New York
Los Angeles
Toronto
London
Sydney
New York
Completed
Ongoing
Pending
Completed
Ongoing
Pending

7. SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Salary > 70000;

EmployeeID	FirstName	LastName	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

8. SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Department IN ('IT', 'Finance');

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

9. SELECT ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE Status <> 'completed';

ProjectID	ProjectName	Budget	Status
102	Employee Training	50000	Ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

10. SELECT ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE Budget > 70000 AND Status <> 'Complete';

ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

11. SELECT EmployeeID, firstName, LastName, Department, Salary, City
 FROM Employees
 WHERE City IN ('New York', 'Toronto')
 ORDER BY Salary DESC;

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

12. SELECT EmployeeID, firstName, LastName, Department, Salary
 FROM Employees
 ORDER BY Salary DESC
 LIMIT 3;

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

13. SELECT Department, SUM(Salary) AS TotalSalary
 FROM Employees
 GROUP BY Department
 ORDER BY TotalSalary DESC;

Department	TotalSalary
Finance	155000
IT	142000
Sales	65000
HR	60000

14. SELECT City, AVG (Salary) AS AverageSalary
FROM Employees
GROUP BY City
HAVING AVG (Salary) > 65000;

City	AverageSalary
New York	75000
Toronto	75000
London	72000

15. SELECT Department, COUNT (EmployeeID) AS EmployeeCount
FROM Employees
GROUP BY Department
HAVING COUNT (EmployeeID) > 1;

Department	EmployeeCount
IT	2
Finance	2

16. SELECT Status, COUNT (ProjectID) AS ProjectCount
FROM Projects
GROUP BY Status
HAVING COUNT (ProjectID) >= 2;

Status	ProjectCount
Completed	2
Ongoing	2
Pending	2

17. SELECT e.EmployeeID, e.FirstName, e.LastName, SUM (p.Budget) AS TotalProjBud
FROM Employees AS e
JOIN Projects AS p ON e.EmployeeID = p.EmployeeID
GROUP BY e.EmployeeID, e.FirstName, e.LastName
HAVING SUM (p.Budget) > 150000;

EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	175000